FINAL PROJECT REPORT

Predicting Parkinson's Disease Using Voice Recordings

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1. Introduction

Parkinson’s disease (PD) is a progressive neurological disorder that affects movement and speech, and early detection is important for better treatment and management. One of the key early signs of PS is a change in voice, with studies showing that about 90% of patients develop some type of vocal impairment, such as loudness, irregular pitch, or breathy speech. These changes happen because the disease affects the muscles that control speech, making voice signals a useful source of information for diagnosis. As highlighted in the journal article, researchers can extract features from voice recordings, such as frequency variation, amplitude changes, and noise levels, and use them in machine learning models to distinguish between individuals with and without PD. This approach is especially valuable because it is non-invasive, low-cost, and can be done remotely, thereby making it suitable for early detection and ongoing monitoring of the disease (Sakar et al., 2019).

This project applies supervised statistical learning methods to classify Parkison’s disease (PD) status using features derived from voice recordings. In this setting, each observation consists of a set of predictor variables (voice features) and a corresponding outcome label indicating whether the individual has PD or not. The goal is to learn a function that maps these predictors to the probability of belonging to the PD class, which makes this a binary classification problem. Supervised learning uses labeled data to learn the relationship between input features and an outcome, and then evaluates how well the model performs on new, unseen data. In this project, models such as logistic regression estimate the probability of Parkinson’s disease (PD), while more flexible methods like tree-based models and support vector machines can capture more complex patterns in the data. Model performance is evaluated using metrics such as accuracy, sensitivity, specificity, and AUC, which together provide a clear understanding of how well the model distinguishes between PD and non-PD individuals, consistent with standard approaches in statistical learning (James et al., 2021).

2. Data and Preprocessing

2.1 Dataset

The dataset consists of voice recordings from 252 subjects, with three recordings per subject, resulting in 756 observations at the recording level. A wide range of features were extracted, including MFCC, wavelet-based, and vocal fold features, which capture key characteristics of speech such as frequency variation, amplitude, and signal complexity.

2.2 Data Aggregation

Because the data are structured at the recording level, each subject contributes three observations that are not independent. To address this, the data were aggregated to the subject level using the subject ID, where the mean of each feature across the three recordings was computed so that each subject is represented by a single observation. The outcome variable was assigned using the first observed class per subject, as it is consistent across recordings.

This aggregation resulted in 201 subjects in the training set and 51 subjects in the test set, confirming that the data were successfully converted from recording-level to subject-level format. No missing values were present after aggregation. This step reduces within-subject variability and avoids biased estimates that could arise from treating repeated measurements as independent observations.

2.3 Data Preprocessing

To improve model performance, several preprocessing steps were applied. First, near-zero variance filtering was performed, although no features were removed at this stage. Next, a correlation analysis of baseline features (see Appendix A) revealed strong multicollinearity, particularly among jitter and shimmer variables. To address this, a high-correlation filter (cutoff = 0.90) was applied, resulting in the removal of 400 highly correlated features, reducing the feature set from 753 to 353 predictors. This step helps stabilize models sensitive to multicollinearity, such as logistic regression (James et al., 2021).

The impact of this filtering is further illustrated in Appendix B, which shows the distribution of feature variance before and after removal, highlighting that many low-information or redundant features were eliminated. After feature selection, all remaining predictors were standardized using z-score normalization, with parameters estimated on the training set only to avoid data leakage. Finally, the outcome variable (PD vs. non-PD) was converted into a categorical factor for classification.

2.4 Exploratory Data Analysis

Exploratory analysis revealed a clear class imbalance in the training data, with approximately 74.6% PD cases and 25.4% non-PD cases (**Figure 1**). This imbalance is important because it can bias models toward predicting the majority class, which highlights the importance of using evaluation metrics such as sensitivity, balanced accuracy, and AUC, rather than relying solely on overall accuracy.

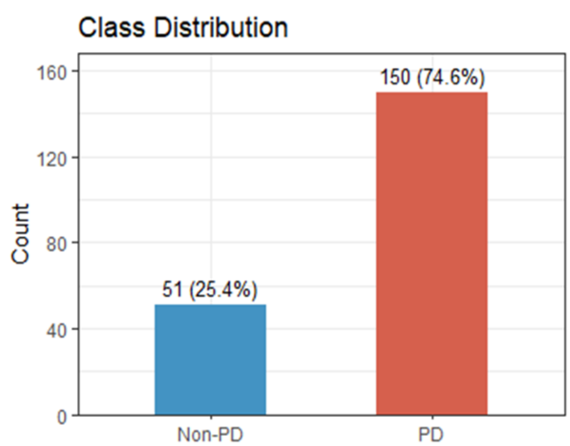


Figure 1: Class Distribution

Figure C (Appendix) illustrates the distribution of selected features by class. Several features, including f1, mean MFCC coefficient, and mean intensity, show noticeable shifts between PD and non-PD groups, indicating their potential usefulness for classification. In contrast, some variables such as GNE_mean and tqwt_energy_dec_1 exhibit highly skewed distributions with limited separation. Overall, while certain features demonstrate clear differences between groups, there is still substantial overlap across most variables, suggesting that no single feature is sufficient to distinguish PD from non-PD. This reinforces the need for models that can combine multiple features to capture complex patterns in the data.

2.5 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was performed for visualization to explore the underlying structure of the data. Figure D (Appendix) shows the projection of observations onto the first two principal components, which explain approximately 14.2% and 11.1% of the variance, respectively. While some separation between PD and non-PD groups is visible, there is substantial overlap, indicating that the classes are not easily separable in lower dimensions. Additionally, the cumulative variance plot (Figure E, Appendix) shows that approximately 36 principal components are required to explain 80% of the variance, indicating that the data are high-dimensional and complex. This suggests that effective classification requires models capable of capturing multivariate relationships, rather than relying on simple linear separation.

3. Methods

3.1 Models

Four supervised classification models were implemented to predict Parkinson's disease (PD) status, using approaches grounded in statistical learning (James et al., 2021). Model training was conducted using 5-fold cross-validation to tune model parameters and improve generalizability, and up-sampling was applied within the training folds to address class imbalance. Two versions of the dataset were used: a correlation-filtered dataset for logistic regression to reduce multicollinearity, and the full feature dataset for more flexible models to preserve potential nonlinear relationships.

Logistic Regression was used as a baseline model due to its interpretability and stability. A penalized logistic regression model (glmnet) was fitted using the filtered dataset, with regularization parameters (α and λ) tuned via cross-validation to handle high-dimensional and correlated predictors.

Random Forest (RF) was applied as an ensemble method, combining predictions from multiple decision trees to improve accuracy and reduce variance. The model was trained on the full feature set, with the number of variables sampled at each split (mtry) tuned using cross-validation.

Support Vector Machine (SVM) with a radial basis function kernel was used to capture nonlinear relationships in the data. Hyperparameters, including the kernel width (σ) and regularization parameter (C), were selected through cross-validation.

Multilayer Perceptron (MLP), a feedforward neural network, was used to model complex nonlinear patterns. To reduce dimensionality, principal component analysis (PCA) was applied prior to modeling, retaining components that explain approximately 80% of the variance. Network size and regularization parameters were tuned using cross-validation.

3.2 Model Evaluation

Model performance was evaluated using several classification metrics, including accuracy, sensitivity, specificity, precision, F1-score, and area under the ROC curve (AUC). While accuracy provides an overall measure of correct predictions, it can be misleading in imbalanced datasets. Therefore, sensitivity and specificity were emphasized to better assess the model's ability to correctly identify PD cases and non-PD cases, respectively. The F1-score provides a balance between precision and recall, while AUC summarizes the model's ability to discriminate between classes across different classification thresholds (James et al., 2021).

To obtain reliable estimates of model performance, 5-fold cross-validation was used, providing a more robust assessment of generalizability than a single train–test split. For logistic regression, regularization methods (ridge and lasso) were implemented using the glmnet framework. Ridge regression ($\alpha = 0$) reduces coefficient variance by shrinking all coefficients, while lasso regression ($\alpha = 1$) performs both shrinkage and variable selection by setting some coefficients to zero. The tuning parameters α and λ were selected through cross-validation to balance bias and variance, improving model performance in the presence of high-dimensional and correlated predictors. For the remaining models, key hyperparameters were similarly tuned using cross-validation.

4. Results

4.1 Model Performance Table

```
##      Model       AUC  Accuracy Sensitivity Specificity Precision        F1
## 1 Logistic 0.8585621 0.8109453   0.8400000   0.7254902 0.9000000 0.8689655
## 2       RF 0.8417647 0.8159204   0.9400000   0.4509804 0.8343195 0.8840125
## 3      SVM 0.7960784 0.8009950   0.9666667   0.3137255 0.8055556 0.8787879
## 4      MLP 0.8162092 0.7562189   0.8000000   0.6274510 0.8633094 0.8304498
```

Table 1 summarizes the performance of all models. While overall accuracy was similar across models (approximately 75–82%), their ability to correctly identify PD and non-PD cases differed substantially. Logistic regression achieved the highest AUC (0.8586) and provided the most balanced performance, with strong sensitivity (0.8400) and the highest specificity (0.7255), indicating effective discrimination between classes. In contrast, Random Forest and SVM achieved very high sensitivity (0.9400 and 0.9667, respectively), meaning they were effective at identifying PD cases, but had much lower specificity (0.4510 and 0.3137), indicating poor identification of non-PD individuals. The MLP model showed moderate performance across all metrics but did not outperform logistic regression. Overall, these results highlight a trade-off between sensitivity and specificity, with logistic regression providing the best overall balance for this classification task.

4.2 ROC Curves

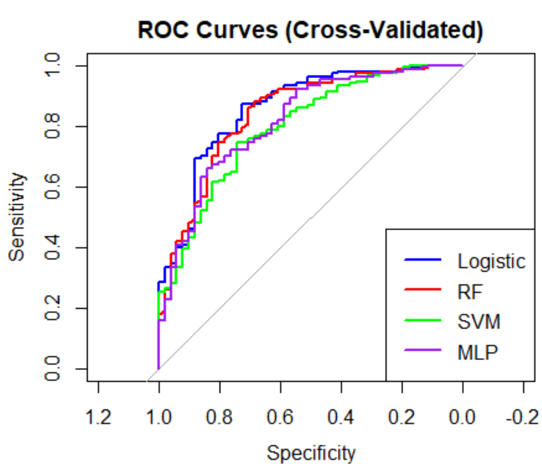


Figure 2: ROC Curves

Figure 2 presents the cross-validated ROC curves for all models. Logistic regression consistently lies above the other models across most thresholds, indicating the highest overall AUC and strongest discrimination between PD and non-PD cases. Random Forest and MLP show comparable performance, closely following the logistic curve, while SVM performs worst overall, with a curve that remains closer to the diagonal reference line. These results suggest that logistic regression provides the most reliable classification across varying thresholds.

4.3 Sensitivity vs Specificity Trade off

The trade-off between sensitivity and specificity across different classification thresholds is shown in Figure F (Appendix). As the threshold increases, sensitivity increases while specificity decreases, illustrating the expected inverse relationship between the two metrics. This highlights the importance of selecting an appropriate threshold based on the application, particularly in clinical settings where sensitivity may be prioritized to reduce the risk of missing PD cases.

4.4 Model Performance (Resampling)

Figure G (Appendix) presents the resampling-based comparison of model performance across key metrics, including sensitivity, specificity, and ROC. Random Forest achieves the highest sensitivity, indicating strong ability to detect PD cases, but this comes at the expense of lower specificity. Logistic regression provides the most balanced performance across all metrics, maintaining relatively high sensitivity while preserving better specificity. In contrast, SVM shows extreme sensitivity but very poor specificity, suggesting a tendency to over-classify observations as PD. Overall, these results reinforce that logistic regression offers the best trade-off between detecting PD cases and avoiding false positives, making it the most reliable model among those considered.

4.5 Feature Importance

Figure H (Appendix) presents the most important predictors identified by the Random Forest model. The top features are primarily related to delta energy measures and wavelet-based (TQWT) features, which capture changes in signal energy and time-frequency characteristics of speech. These results are consistent with existing literature, which shows that wavelet-based features and MFCC-related measures are effective in capturing vocal impairments associated with Parkinson’s disease. The importance of these features suggests that variations in speech energy and frequency patterns play a key role in distinguishing PD from non-PD individuals.

5. Discussion

5.1 Key Findings

Logistic regression achieved the best overall performance, with the highest AUC and a strong balance between sensitivity and specificity. This suggests that the penalized model was able to handle high-dimensional and correlated predictors effectively while maintaining stability. In contrast, Random Forest and SVM achieved very high sensitivity but had extremely low specificity, indicating that these models tended to classify most observations as PD. The MLP model, which used PCA for dimensionality reduction, showed moderate performance but did not outperform logistic regression.

5.2. Interpretation

These results reflect the bias-variance trade-off. More flexible models, such as Random Forest and SVM have low bias but higher variance, which may lead to overfitting, particularly in imbalanced data. Logistic regression, especially with ridge regularization, introduces some bias but reduces variance, resulting in better generalization. Additionally, accuracy alone is not sufficient in imbalanced data sets, and metrics such as sensitivity and specificity provide a more meaningful evaluation. The comparable or superior performance of logistic regression suggests that the underlying relationship between features and PD status is moderately linear, and does not require highly flexible models.

5.3. Clinical Implications

From a clinical perspective, missing PD cases (false negatives) is particularly concerning. While models such as RF and SVM maximize sensitivity, they do so at the expense of very low specificity, leading to a high number of false positives. Logistic regression provides the best overall trade-off, making it more suitable for practical screening scenarios where both accurate detection and reliable classification are important.

6. Limitations

This study has several limitations. First, the class imbalance in the dataset may have influenced model behavior, leading some models to favor the majority PD class. Second, although hyperparameters were tuned using cross-validation, extensive optimization was not performed, which may limit the full potential of certain models. In particular, the neural network (MLP) was not extensively tuned in terms of architecture or parameters due to time and computational constraints, and because the relatively small sample size increases the risk of overfitting with more complex configurations. Finally, external validation was not conducted, so the generalizability of the models to independent datasets remains uncertain.

7. Conclusion

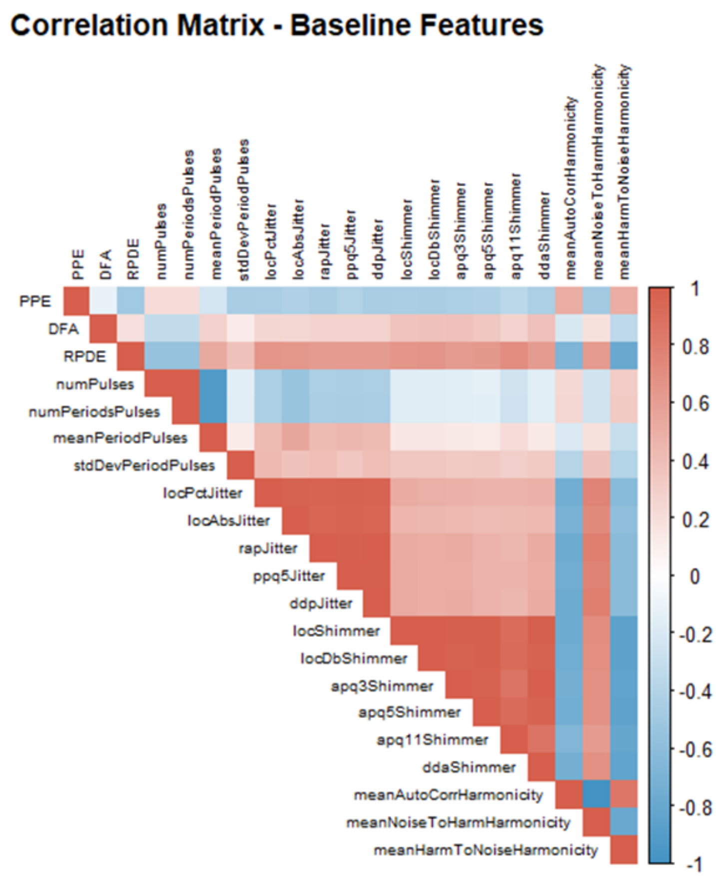
This study demonstrates that voice-based features can effectively be used to detect Parkinson’s disease, highlighting the potential of non-invasive, data-driven approaches for early screening. While more complex models such as Random Forest and SVM achieved high sensitivity, they performed poorly in identifying non-PD cases. In contrast, logistic regression provided the best overall balance between interpretability, sensitivity, and specificity, making it more reliable for practical use. Overall, these findings emphasize that simpler, well-regularized models can perform as well as or better than more complex approaches, and that model selection should be guided by both predictive performance and clinical relevance.

8. References

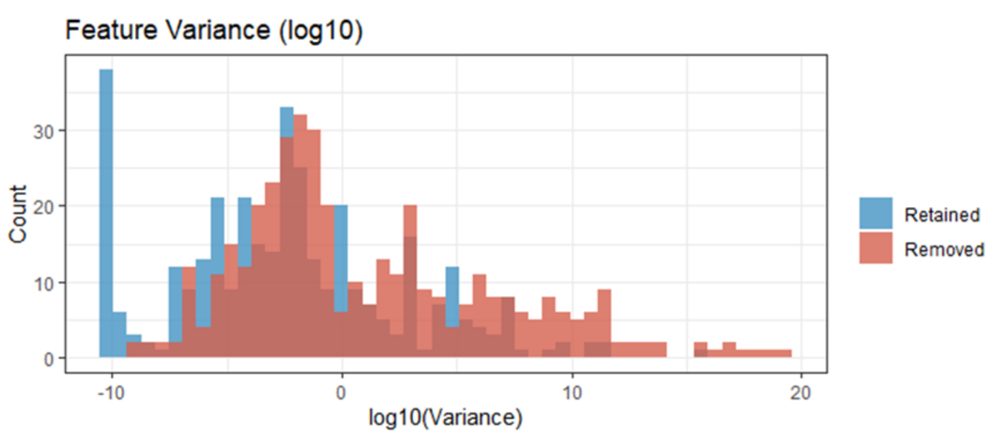
- Sakar, B. E., Isenkul, M. E., Sakar, C. O., Sertbas, A., Gurgen, F., Delil, S., Apaydin, H., & Kursun, O. (2019). **Collection and analysis of a Parkinson speech dataset with multiple types of sound recordings**. *IEEE Journal of Biomedical and Health Informatics, 23*(5), 2146–2154. https://doi.org/10.1109/JBHI.2018.2878022
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in R* (2nd ed.). Springer. https://doi.org/10.1007/978-1-0716-1418-1

9. List of Appendices

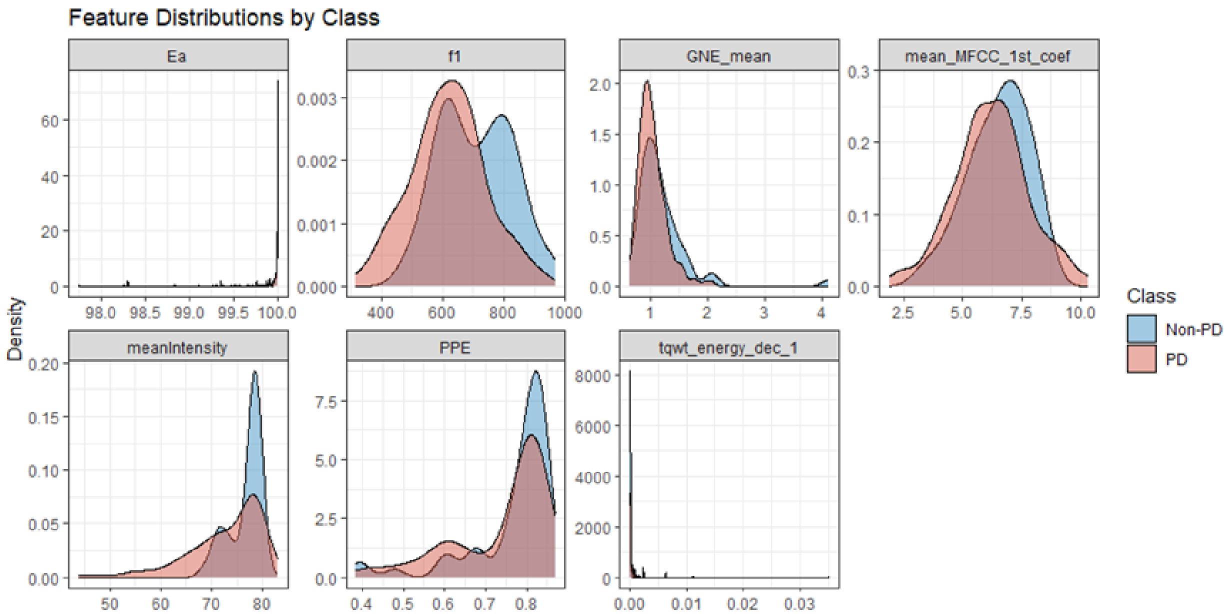
9.1 Appendix A: Correlation Matrix (Baseline Features)



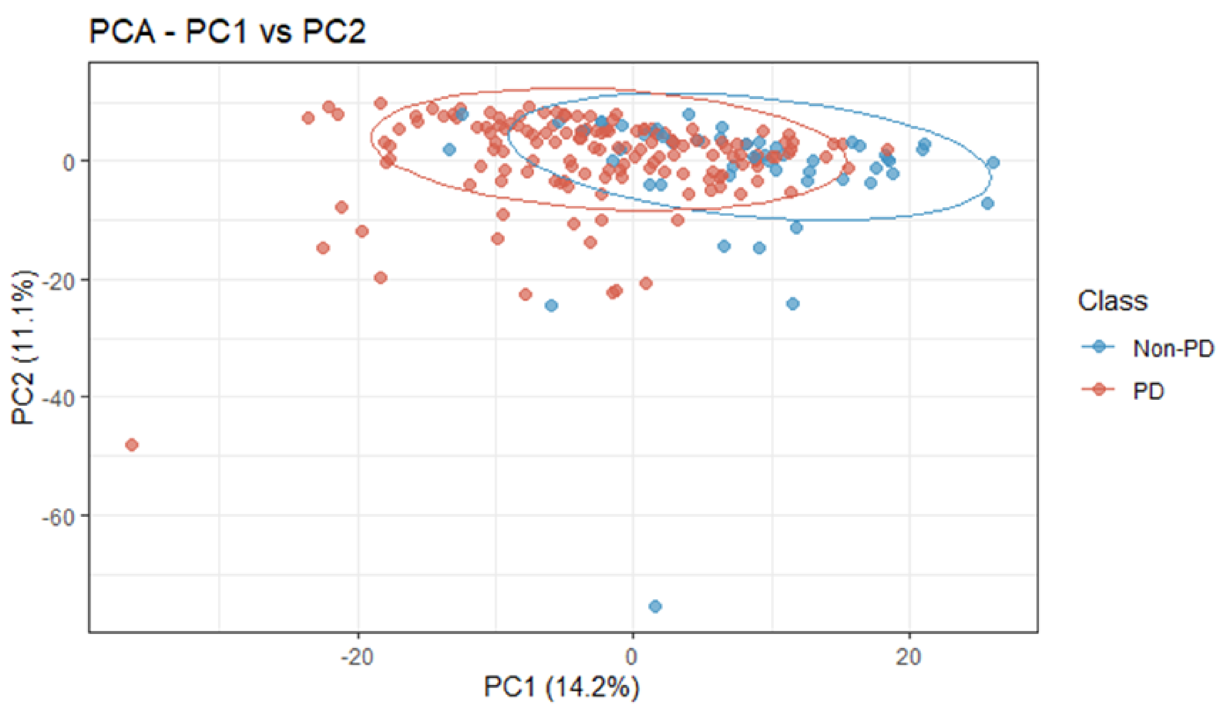
9.2 Appendix B: Variance Distribution



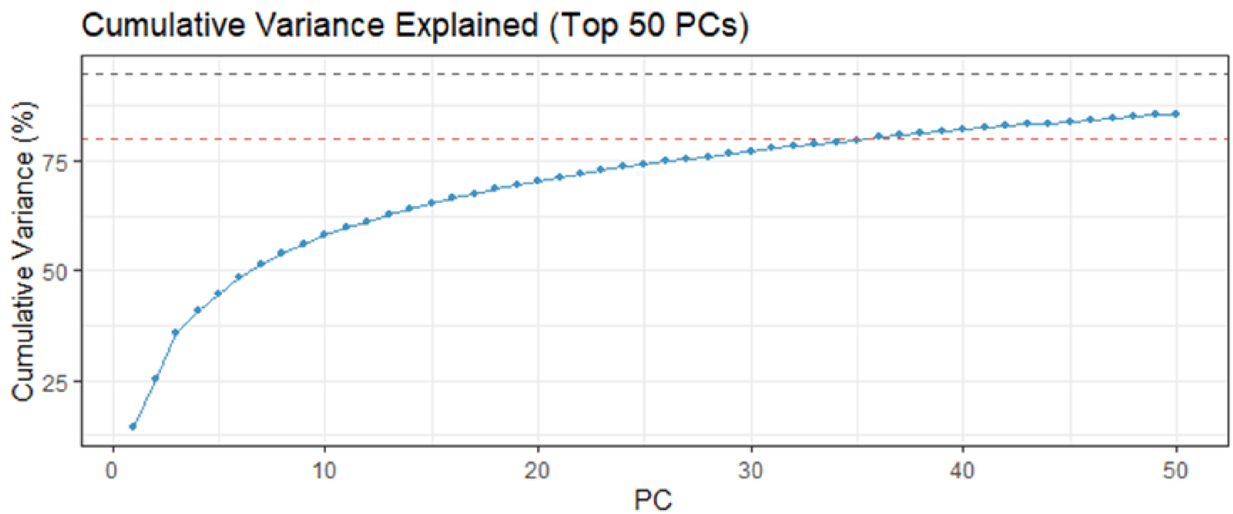
9.3 Appendix C: Feature Distributions By Class



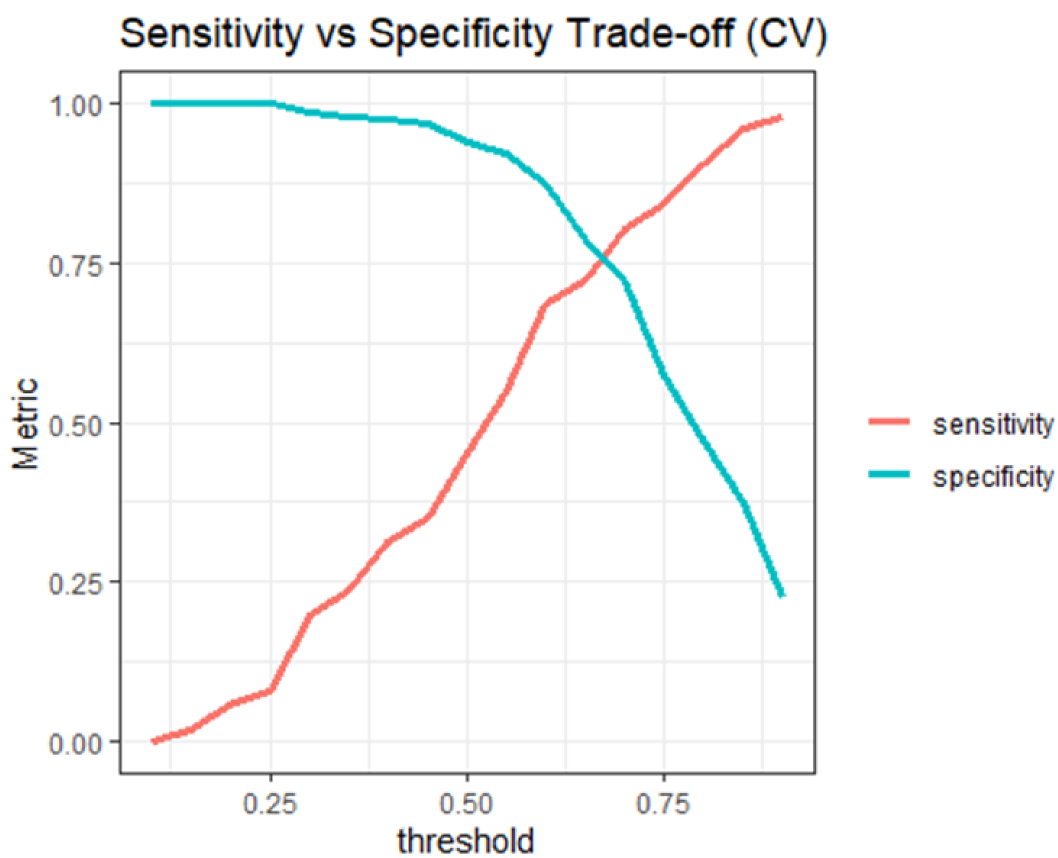
9.4 Appendix D: Principal Component Analysis (PCA) Scatter Plot



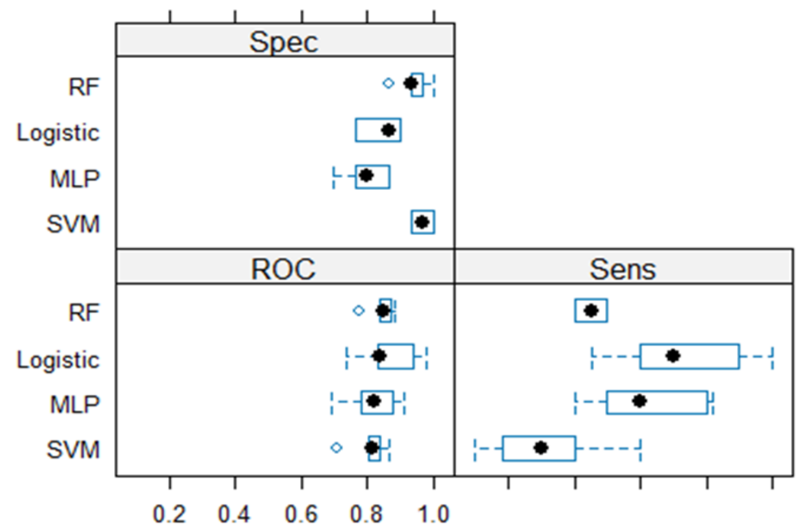
9.5 Appendix E: Cumulative Variance Plot



9.6 Appendix F: Sensitivity vs. specificity trade-off across classification thresholds. Increasing the threshold improves sensitivity at the cost of reduced specificity.



9.6 Appendix G: Model comparison based on resampling (5-fold cross-validation), showing variability in sensitivity, specificity, and ROC across folds.



9.7 Appendix H: Feature importance plot from the Random Forest model, showing the top predictors contributing to classification performance.